

Generalizing Visual Proprioception for Multiple Robot Configurations

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Abstract—Future robotic systems will likely consist of a wide variety of robot embodiments, operating in unpredictable environments, with compliant joints and casual use of tools for which no precise models are available. Visual proprioception is a promising approach that can complement the limited internal proprioception capabilities of such robots. However, a key question is how generalizable visual proprioception is across different embodiments. In this work we investigate the generalization of visual proprioception across three very different robot configurations: a low-cost and low-precision 6DOF AL5D arm, a high precision 7DOF Franka arm, and a humanoid G1 arm with 7DOF. We investigate whether the same proprioception model architecture can be used with different robots with minimal changes (e.g. changing the output size of the regression model). Furthermore, we investigate whether parts of actual model, such as the pretrained vision encoder can be reused across different robots.

I. INTRODUCTION

While science fiction movies often feature futures with shiny teams of identical humanoid robots, the reality is that we will likely have a wide variety of robot embodiments, determined by cost, age, wear-and-tear, and the specific tasks they are designed for. As robots move out from the controlled environments of factories and warehouses into the real world, they often need to adapt to the unpredictable external environment as well as the limitations and uncertainties of their own internal sensors and actuators. This is especially true for robots that are designed to be low cost, motion-compliant as they may not have high precision proprioception capabilities.

Visual proprioception, which uses visual information to infer the state of the robot, is a promising approach to address these challenges. It can be applied to both the robot and the tools it uses, and it can be relatively cheap compared to other sensing modalities. However, a key question is how generalizable visual proprioception is across different embodiments. In particular, we are interested in situations where the kinematic structure of the robots is different, and even when the configuration space of the robots is of different sizes.

In this work, we build on the visual proprioception mechanism of [1] and assume a direct flow approach, where visual observations are used by a *vision encoder* to create a latent embedding of moderate size (128D or 256D), which is then

used by a *regression model* to infer the configuration of the robot. We consider three very different robot configurations: a low-cost and low-precision 6DOF AL5D arm, a high precision 7DOF Franka arm, and a humanoid G1 arm with 7DOF. We investigate whether the same model architecture can be used with different robots with minimal changes (e.g. changing the output size of the regression model). Furthermore, we investigate whether parts of actual model, such as the pretrained vision encoder can be reused across different robots.

Figure 1 shows the proposed experiment flow. For all robot configurations, the input is the figure from a fixed camera that captures the robot and its workspace. The input is fed into a vision encoder, which produces a latent embedding. The embedding is then fed into a regression model, which produces the estimated configuration of the robot. These proprioception models are tested against different other robot configurations.

II. RELATED WORK

Visual proprioception, recovering a robot’s joint state from camera observations, has been approached from several distinct angles.

Latent-space regression. A number of methods first encode the input image into a learned feature vector and then regress joint angles from that representation [1]–[3]. Widmaier et al. [2] take a per-pixel approach, training a random regression forest on features extracted from segmented depth images to predict joint angles on a frame-by-frame basis. Liu et al. [3] propose RoboTAG, which frames configuration estimation as an end-to-end problem by constructing a topological alignment graph that captures the spatial relationships between robot links, removing the need for separate keypoint detection. Goswami et al. [4] leverage self-supervised embedding pre-training to jointly estimate robot pose and joint angles from RGB input without requiring rendered training data.

Keypoint-based methods. Rather than regressing directly from a global feature, several works first localize the robot joints and then solve the robot configuration geometrically [5]–[7]. Ortenzi et al. [5] track multiple robot segments with a monocular camera and recover the kinematics through transformation-matrix optimization, in a later work [8] they boosted accuracy by attaching fiducial markers to each joint. Zuo et al. [6] detect 2D keypoints with a stacked hourglass network and convert them to 3D via perspective projection, training entirely on synthetic renderings. Tian et al. [7] follow a similar detect then convert strategy but incorporate the robot’s structural constraints during the 2D to 3D reconstruction step.

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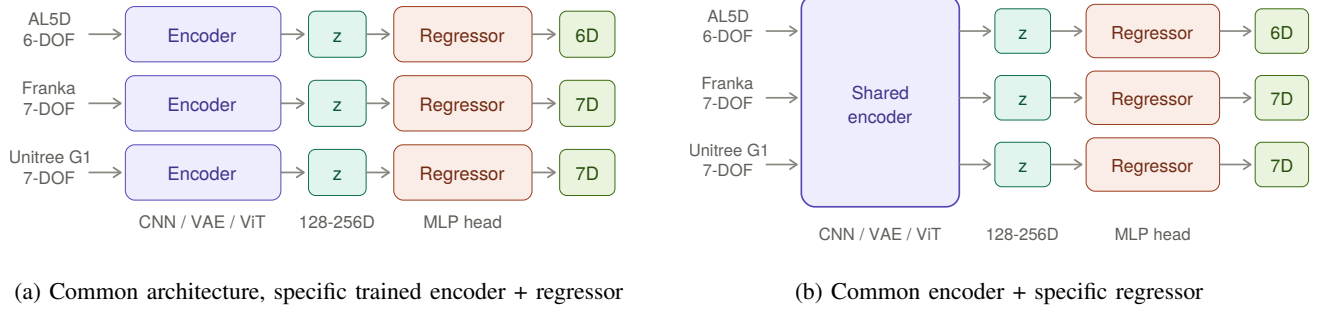


Fig. 1: The architecture of the proposed visual proprioception mechanism. (a) The same encoder-regressor architecture is trained independently for each robot, with only the output dimension changing to match the robot’s degrees of freedom. (b) A single shared encoder is trained once and frozen, while a lightweight regressor is fitted per robot, enabling cross-platform transfer with minimal data.

Render-and-compare. An alternative family of approaches maintains an internal 3D model of the robot and adjusts its rendered pose until it matches the target image. Labbé et al. [9] for example starts from an initial guess, then repeatedly renders configurations and refines the estimate, achieving strong accuracy on self-occluded joints.

Camera-to-robot calibration. A task closely related to proprioception is estimating the extrinsic pose of the camera relative to the robot. Lee et al. [10] train a keypoint detector on domain-randomized synthetic images and recover the camera pose through Perspective-n-Point solvers; their accompanying DREAM dataset, which includes Franka Panda sequences, serves as one of our evaluation benchmarks. Lu et al. [11] replace the keypoint pipeline with a fully differentiable renderer, achieving comparable speed.

Sim-to-real transfer and large-scale learning. Bridging the gap between simulation and the real world is central to scaling visual robot control. He et al. [12] train a teacher policy in simulation for humanoid locomotion and manipulation, then convert it into a vision-based student network using aggressive visual domain randomization. Garcia et al. [13] demonstrate that randomizing textures, lighting, and object appearance across seven manipulation tasks is sufficient to deploy a UR5 policy with zero real-world fine-tuning. Levine et al. [14] on the other hand collect hundreds of thousands of real grasp attempts to learn hand-eye coordination, highlighting both the potential and the cost of purely real-data approaches.

Our work differs from the methods above in its emphasis on generalization across structurally diverse platforms. Rather than tailoring the pipeline to a single robot, we evaluate whether the same encoder-regressor architecture can serve a low-cost 6DOF arm, an industrial 7DOF manipulator, and a humanoid limb, and we investigate cross-robot transfer where the encoder is trained on one platform and the regressor is fitted on another.

III. METHODOLOGY

Figure 2 illustrates the three robotic platforms used in our experiments: the 6DOF AL5D arm captured in a real-world setting (Figure 2c), the 7DOF Franka Panda drawn from the

DREAM dataset [10] (Figure 2b), and the 7DOF left arm of the Unitree G1 humanoid (Figure 2a). For every platform, we gathered 10 recording sessions of approximately 500 motions each. Four sessions were reserved for encoder training, another four for fitting the proprioception regressor, and the remaining two served as held-out validation sets.

A. Problem Setup and Models

A robot’s pose is encoded as a vector $\mathbf{a} \in [0, 1]^d$ with $d \in \{6, 7\}$, with the goal of minimizing θ .

$$\theta^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \text{dist}(\mathbf{a}_i, f(\mathbf{z}_i; \theta)). \quad (1)$$

The regressor itself is a compact MLP: a 128 or 256-dimensional input layer, two 64-unit hidden layers with ReLU activations, and an output layer whose width matches the robot’s degrees of freedom. Each regressor is trained independently with early stopping governed by validation loss. When an encoder requires supervised fine-tuning, a temporary copy of this MLP provides the gradient signal and is discarded once the encoder weights are frozen.

Because the system is intended for real-time operation on resource-constrained hardware, we require the encoder to complete a forward pass in tens of milliseconds. Computationally expensive architectures are therefore permitted only during offline training, not at deployment.

B. Encoder Architecture Variants

We compare three families of visual encoders, each projecting the input image to a 128 or 256-dimensional latent vector $\mathbf{z}$.

Convolutional VAE. A convolutional variational autoencoder [15] is trained in a fully unsupervised manner on frames of the robot moving against a fixed background. Because the robot is the dominant source of visual change, the learned latent space is implicitly biased toward capturing its pose. At inference time we retain only the encoder’s mean output $\mathbf{z} = \mu(o)$, dropping both the stochastic sampling layer and the decoder.



Fig. 2: Visuals of each of the robotic arms used in the experiments

Fine-tuned CNN backbone. Off-the-shelf feature extractors such as VGG-19 [16] and ResNet-50 [17] yield representations far larger than what proprioception demands, and much of their capacity encodes information irrelevant to robot state. We address this by appending a small MLP that projects the backbone features down to a 128 or 256-dimensional bottleneck. This projection layer is optimized end-to-end on a labeled proprioception set, using an auxiliary regressor head to supply the training objective. Once fine-tuning is complete, we retain only the projection; the bottleneck activations become z .

Vision Transformer backbone. Vision Transformers (ViT) [18] offer an alternative to convolutional encoders, with particular strength in capturing long-range spatial dependencies across the image. However, pretrained ViT models produce feature vectors that are substantially wider than our target latent size. For instance, ViT-Base outputs a 768-dimensional representation. To bridge this gap, we attach a multi-layer projection network composed of fully connected layers with batch normalization and dropout, progressively reducing the feature dimension down to 128 or 256. As with the CNN encoders, this projection is trained using a supervised proprioception objective through a temporary regressor head, which is discarded after the projection weights have converged. The frozen transformer backbone handles feature extraction while only the lightweight projection layers are updated during fine-tuning.

IV. EXPERIMENTAL RESULTS

In this section we present the results of a series of experiments evaluating the performance of the visual proprioception framework across three different robotic platforms as well as the transferability of the learned encoders across platforms.

A. Transferring the architecture across different robotic arms

In the first series of experiments, we use the same proprioception architecture over three different robotic platforms, with different number of degrees of freedom, kinematics, and visual appearances. In these experiments, no training weights

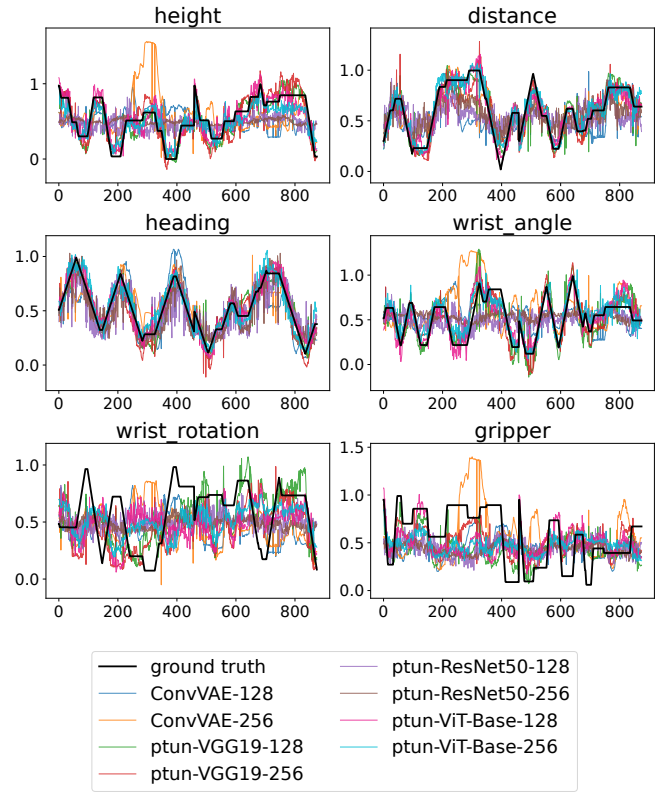


Fig. 3: The proprioception estimates compared to the ground truth for the real AL5D robotic arm. The graphs track the evolution in time of the configuration components while the robot performs a randomly generated movement designed to cover a large part of the configuration space.

were transferred across platforms, basically showing the case of Figure 1a.

Figure 3 shows the results for the real-world AL5D robotic arm. In the case of this arm, the chosen configuration components are the height, distance, heading, wrist angle, wrist rotation and gripper opening. While the latter three have a direct mapping to the end-effector position of the corresponding joint, the first three are calculated from joint

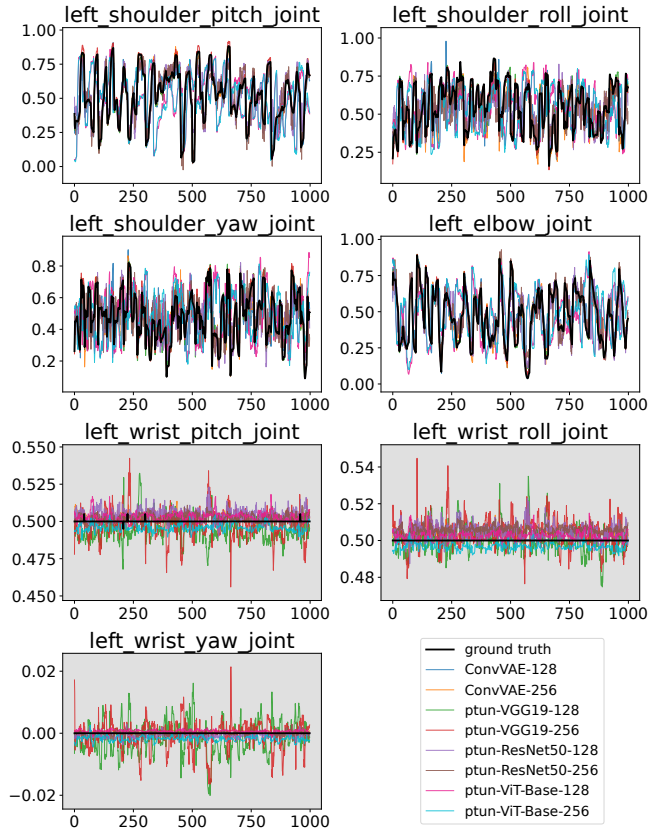


Fig. 4: Results for the proprioception of a sim 7DOF Unitree G1 robotic arm (gray background non-moving joint) (Best viewed in color)

values using forward kinematics. We find that different encoders perform at different performance levels across the different configuration components, however, in general most encoders perform well on the heading and distance, which are the most visually salient configuration components for this arm. On the other hand, the gripper opening and wrist rotation are more difficult to estimate, which can be attributed to the fact that they have a smaller visual signature and they are often occluded during the arm’s movement.

Figure 4 shows results for the arm of Unitree G1 humanoid robot. The elements of the configuration vector for this arm are the 7 joint angles of the left arm. The robot was simulated in Isaac Sim [19]. On the trajectories simulated in the validation dataset only 4 joints are moving, while the other 3 are fixed (and grayed out for easier identification on the figure). We find that the proprioception models track the moving joints well, while exhibiting some degree of noise on the non-moving joints.

Figure 5 shows results for the Franka Panda robotic arm. The configuration vector for this arm is also composed of the 7 joint angles. The validation dataset is taken from real world robot data from [10]. Similarly to the Unitree G1, only 6 joints are moving in the validation dataset, while one joint is fixed (and grayed out for easier identification on the figure). Overall, we find that the proprioception models work

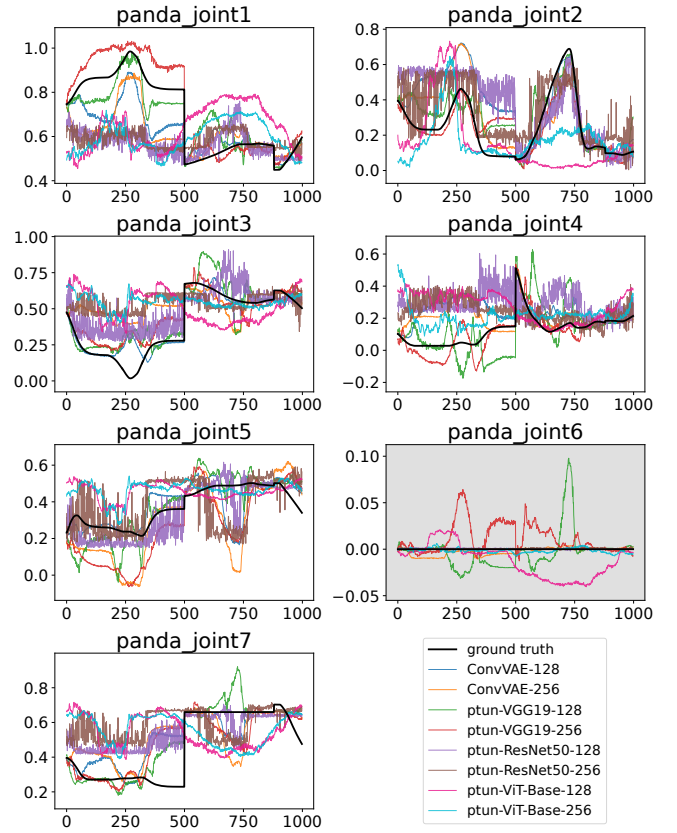


Fig. 5: Results for the proprioception of a real 7DOF Franka Panda robotic arm (gray background non-moving joint) (Best viewed in color)

less well on this dataset compared to the previous two.

B. Encoder transfer across robotic arms

In the second series of experiments, we evaluate the transferability of the learned encoders across platforms. In these experiments, the encoder is trained on one platform and then frozen, while only the regression model is trained on a different platform. This corresponds to the case of Figure 1b.

Figure 6 shows the results for the case of training the encoder on the simulated Unitree G1 robotic arm and using this encoder to provide the input to the regression model for the real Franka Panda robotic arm. We find that the performance of the proprioception models is overall worse than when training the encoder directly on the target platform, however, it still tracks well some of the joints, especially those with a strong visual signature. This suggests that the learned encoder is able to extract some of the visual features that are relevant for proprioception across different platforms, and that only a small amount of data is needed to fit the regression model to a new platform.

Figure 7 shows the results for the opposite case of training the encoder on the real Franka Panda robotic arm and using this encoder to fit a regression model for the simulated Unitree G1 robotic arm.

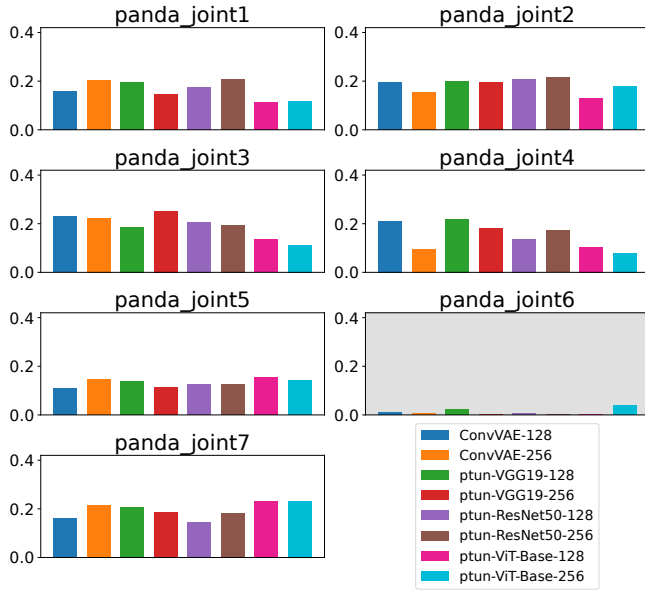


Fig. 6: Joint-wise proprioception MSE for a Franka Panda robotic using an encoder trained on a Unitree G1 robotic arm (gray background non-moving joint)

C. Investigating the best visual encoder

One of the interesting conclusions from the previous experiments is that no single encoder architecture dominates across all platforms and configuration components. Table I shows the best performing encoder for each configuration component of each platform, as well as the corresponding MSE. We find that ViT-Base is the overall best-performing architecture, winning 12 out of 26 moving joints (46%) across all configurations. However, the winner varies by platform: VGG19 dominates on the simulated G1 (4/4 moving joints), ViT-Base on the real-world AL5D (4/6) and Franka Panda (4/6) and cross-robot G1→Franka transfer (4/6), and ConvVAE on the cross-robot Franka→G1 transfer (4/4). This suggests that the choice of encoder should be guided by robot type and environment rather than assumed to be universal.

D. Training Data Requirements for Regression

Performing a transfer of the pre-trained encoder to a new robotic arm allows us to significantly reduce the amount of training data needed. However, some amount of labeled data is still needed to fit the regression model to the new platform. We have investigated how the performance of the regression model changes as a function of the amount of training data available. Figure 8 shows the MSE results for different models on each joint, while X axis presents the number of images used for training. We find that the MSE decreases significantly as the number of training images increases up to around 300 images, after which the improvements become more marginal. This suggests that while some amount of data is needed to fit the regression model, it does not need to be very large, making this approach practical for rapid deployment on new platforms.

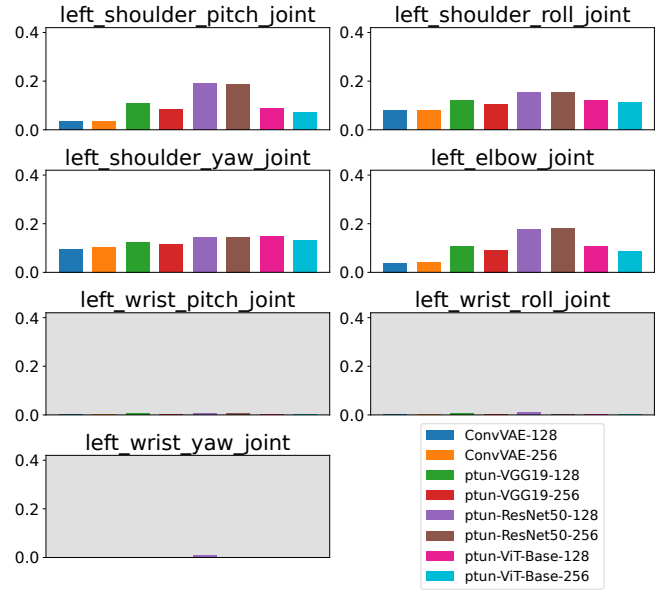


Fig. 7: Joint-wise proprioception MSE for a simulated Unitree G1 robotic arm using an encoder trained on a real Franka Panda robotic arm (gray background non-moving joints)

E. Overall conclusions

No single encoder dominates all scenarios: ViT-Base is the overall best-performing architecture, winning 12 out of 26 moving joints (46%) across all configurations. However, the winner varies by platform: VGG19 dominates on the simulated G1 (4/4 moving joints), ViT-Base on the real-world AL5D (4/6) and Franka Panda (4/6) and cross-robot G1→Franka transfer (4/6), and ConvVAE on the cross-robot Franka→G1 transfer (4/4). This suggests that the choice of encoder should be guided by robot type and environment rather than assumed to be universal.

Cross-robot transfer is feasible: Training the encoder on one robot and fitting only the regressor on another yields reasonable accuracy, making this approach viable when data for a target platform is scarce.

ViT excels on real-world data: ViT appears to generalize better to real images and across robots, as evidenced by its better performance on both the AL5D and Franka Panda as well as the G1→Franka transfer setting.

Visually moving joints are easiest to estimate: Across all robots, joints whose motion produces large visual changes such as height for the AL5D, panda_joint1 for the Franka, and shoulder pitch for the G1 consistently achieve the lowest RMSE.

Moderate dataset sizes suffice: Our dataset-size ablation Figure 8 shows that regression accuracy plateaus at approximately 300 images per recording session. This implies that as few as 1200 labeled image-action pairs are sufficient for effective proprioception on a new platform, making the approach practical for rapid deployment.

Compact latent representation: The latent space dimensionality has a limited effect. Across all encoders, 128-dimensional and 256-dimensional variants perform compa-

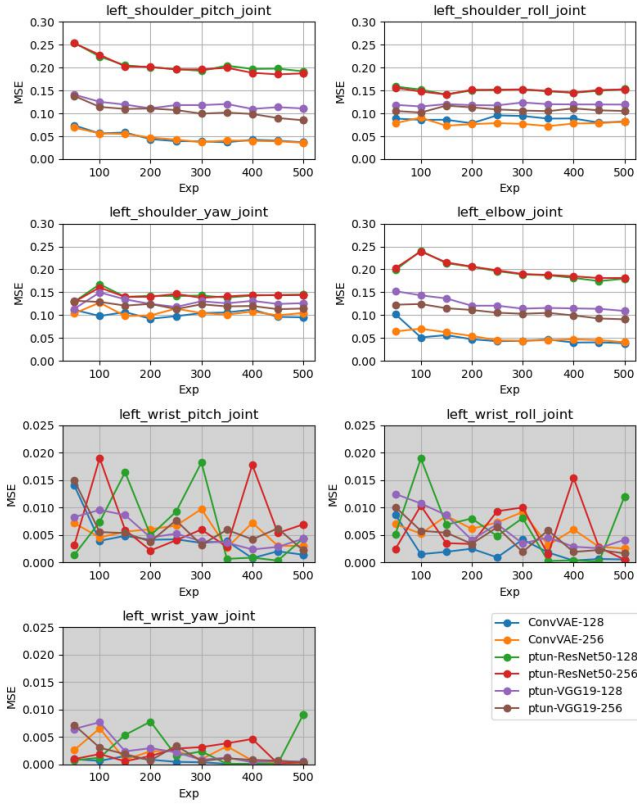


Fig. 8: The accuracy of proprioception for the function of the number of images in the regression dataset for Franka→G1 transfer (non-moving joints have a gray background).

rably, with the 128-dimensional models frequently achieving the best per-joint results.

V. CONCLUSIONS

In this paper we considered the problem of generalizing visual proprioception across different robot embodiments. We found that single pass, direct flow visual proprioception that uses an encoder and a regression model can be effective across three very different robot configurations, both in real-world and simulation settings. However, we also found that no single encoder dominates across all scenarios. Thus, whenever feasible, additional performance can be gained adapting the encoder to the specific robot.

We also found that cross-robot transfer of the pre-trained encoder is also feasible. With such a transfer of only a very small amount of data is needed to fit the regression model to a new robot, making this approach viable when data for a target platform is scarce.

Finally, we found that visually moving joints are easier to estimate than those whose motion produces smaller visual changes. This suggests that future work could investigate how to better capture the proprioception signal from joints with small visual signatures, by incorporating additional sensing modalities or by designing encoders that are more sensitive to subtle visual cues.

Robot	Joint	Best model	RMSE
AL5D	height	ViT-Base-256	0.113
	distance	ViT-Base-128	0.077
	heading	ViT-Base-128	0.062
	wrist_angle	ViT-Base-256	0.096
	wrist_rotation	ResNet50-256	0.245
	gripper	VGG19-256	0.230
	Winner: ViT-Base (4/6 moving joints)		
Unitree G1	left_shoulder_pitch_joint	VGG19-128	0.014
	left_shoulder_roll_joint	VGG19-256	0.022
	left_shoulder_yaw_joint	VGG19-256	0.024
	left_elbow_joint	VGG19-256	0.013
	left_wrist_pitch_joint	ConvVAE-128	0.000
	left_wrist_roll_joint	ConvVAE-128	0.000
	left_wrist_yaw_joint	ConvVAE-128	0.000
Winner: VGG19 (4/4 moving joints)			
Franka Panda	panda_joint1	ViT-Base-128	0.058
	panda_joint2	ViT-Base-128	0.051
	panda_joint3	ViT-Base-128	0.074
	panda_joint4	ViT-Base-256	0.049
	panda_joint5	ResNet50-128	0.079
	panda_joint6	ConvVAE-128	0.001
	panda_joint7	VGG19-128	0.103
Winner: ViT-Base (4/6 moving joints)			
G1 → Franka	panda_joint1	ViT-Base-128	0.113
	panda_joint2	ViT-Base-128	0.131
	panda_joint3	ViT-Base-256	0.113
	panda_joint4	ViT-Base-256	0.079
	panda_joint5	ConvVAE-128	0.108
	panda_joint6	VGG19-256	0.000
	panda_joint7	ResNet50-128	0.147
Winner: ViT-Base (4/6 moving joints)			
Franka → G1	left_shoulder_pitch_joint	ConvVAE-256	0.036
	left_shoulder_roll_joint	ConvVAE-256	0.082
	left_shoulder_yaw_joint	ConvVAE-128	0.095
	left_elbow_joint	ConvVAE-128	0.039
	left_wrist_pitch_joint	ConvVAE-128	0.001
	left_wrist_roll_joint	ResNet50-256	0.001
	left_wrist_yaw_joint	ConvVAE-128	0.000
Winner: ConvVAE (4/4 moving joints)			
Overall winner: ViT-Base			

TABLE I: Investigating the best visual encoder for proprioception accuracy for the three robot embodiments and for two scenarios of model transfer. **Bold**: the configuration component with the best accuracy. Gray: non-moving joint. The winner is the visual encoder that is the best across the most number of configuration components.

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